Experiment ‘p_6_1_c_process_array_reverse_order_seq’ Results

June 5, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_6_1_c_process_array_reverse_order_seq
Natural language statement: null
Method signature: p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<int>)

Ensures

- `|reversed| == |inputs|`
- `forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]`

Requires

- `|inputs| == 10`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<int>)

The method should respect the following contract:

requires |inputs| == 10, ensures |reversed| == |inputs|, ensures forall i :: 0 <= i < |
    ↪ reversed| ==> reversed[i] == inputs[|inputs| - 1 - i]

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnNdvxOq1t3s08drcCgu5pfDcXrrm
Model: gpt-5.5-2026-04-23
Created at: 1780661327
Finish reason: stop
Usage: 338 tokens in, and 907 tokens out

```
{
  reversed := [inputs[9], inputs[8], inputs[7], inputs[6], inputs[5], inputs[4], inputs
      ↪ [3], inputs[2], inputs[1], inputs[0]];

  assert forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i] by {
    forall i | 0 <= i < |reversed|
      ensures reversed[i] == inputs[|inputs| - 1 - i]
    {
      if i == 0 {
        assert |inputs| - 1 - i == 9;
      } else if i == 1 {
        assert |inputs| - 1 - i == 8;
      } else if i == 2 {
        assert |inputs| - 1 - i == 7;
      } else if i == 3 {
        assert |inputs| - 1 - i == 6;
      } else if i == 4 {
        assert |inputs| - 1 - i == 5;
      } else if i == 5 {
        assert |inputs| - 1 - i == 4;
```

```
    } else if i == 6 {
      assert |inputs| - 1 - i == 3;
    } else if i == 7 {
      assert |inputs| - 1 - i == 2;
    } else if i == 8 {
      assert |inputs| - 1 - i == 1;
    } else {
      assert i == 9;
      assert |inputs| - 1 - i == 0;
    }
  }
 }
}
```

Final Program

```
method p_6_1_c_process_array_reverse_order_seq(inputs: seq<int>) returns (reversed: seq<
   ↪ int>)
    requires |inputs| == 10
    ensures |reversed| == |inputs|
    ensures forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i
       ↪ ]
{
 reversed := [inputs[9], inputs[8], inputs[7], inputs[6], inputs[5], inputs[4], inputs
    ↪ [3], inputs[2], inputs[1], inputs[0]];

 assert forall i :: 0 <= i < |reversed| ==> reversed[i] == inputs[|inputs| - 1 - i] by {
  forall i | 0 <= i < |reversed|
   ensures reversed[i] == inputs[|inputs| - 1 - i]
  {
   if i == 0 {
    assert |inputs| - 1 - i == 9;
   } else if i == 1 {
    assert |inputs| - 1 - i == 8;
   } else if i == 2 {
    assert |inputs| - 1 - i == 7;
   } else if i == 3 {
    assert |inputs| - 1 - i == 6;
   } else if i == 4 {
    assert |inputs| - 1 - i == 5;
   } else if i == 5 {
    assert |inputs| - 1 - i == 4;
   } else if i == 6 {
    assert |inputs| - 1 - i == 3;
   } else if i == 7 {
    assert |inputs| - 1 - i == 2;
   } else if i == 8 {
    assert |inputs| - 1 - i == 1;
   } else {
    assert i == 9;
```

```
      assert |inputs| - 1 - i == 0;
    }
  }
 }
}
```

Total Token Usage

Input tokens: 338
Output tokens: 907
Reasoning tokens: 512
Sum of 'total tokens': 1245

Experiment Timings

Overall Experiment started at 1780661326850, ended at 1780661369755, lasting 42905ms (42.91 seconds)
Iteration #1 started at 1780661326875, ended at 1780661369754, lasting 42879ms (42.88 seconds)